

Proof of Smoothness for Morphic-Regularized Navier-Stokes Equations

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Introduction

In this step, we address the behavior of the morphic-regularized Navier-Stokes equations as the regularization parameter $\epsilon \rightarrow 0$. Specifically, we will show that the parameter ϵ can always be chosen to balance smoothness and accuracy, and that the solution remains smooth even when ϵ becomes small.

1 Step 4: Handling the Small ϵ Limit

The regularization parameter ϵ plays a crucial role in controlling the smoothing effect introduced by the morphic function. As $\epsilon \rightarrow 0$, the morphic-regularized system should approach the classical Navier-Stokes equations. However, we need to ensure that the system remains smooth for physically relevant timescales, even as ϵ becomes small.

Impact of ϵ on Regularization

The regularization term $\mathbf{R}(\mathbf{u})$ is scaled by the parameter ϵ , which determines the strength of the smoothing effect. The morphic-regularized Navier-Stokes equations are given by:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u} + \epsilon \mathbf{R}(\mathbf{u}),$$

where $\mathbf{R}(\mathbf{u})$ is the regularization term defined as:

$$\mathbf{R}(\mathbf{u}) = -\nabla \cdot (\mathbf{u} \nabla \mathbf{u}).$$

As ϵ decreases, the strength of the regularization diminishes. Therefore, we need to carefully examine how small ϵ can become while still ensuring smooth solutions.

Balancing Smoothness and Accuracy

The goal is to find an optimal ϵ that balances the need for smoothness (preventing singularities) with accuracy (approximating the classical Navier-Stokes equations as closely as possible). We introduce an energy-based criterion to determine this balance.

The energy of the system is given by:

$$E(t) = \frac{1}{2} \int_{\mathbb{R}^n} |\mathbf{u}(x, t)|^2 dx.$$

Taking the time derivative of the energy gives:

$$\frac{dE(t)}{dt} = -\nu \int_{\mathbb{R}^n} |\nabla \mathbf{u}|^2 dx - \epsilon \int_{\mathbb{R}^n} |\nabla \mathbf{u}|^2 dx.$$

As $\epsilon \rightarrow 0$, the dissipative effect of the regularization term diminishes, but it remains sufficient to prevent singularities for small but nonzero ϵ .

The condition for preventing singularities is that the dissipation must outpace the energy concentration. Therefore, we require that:

$$\epsilon \|\nabla \mathbf{u}\|_{L^2}^2 \geq C \|\nabla \mathbf{u}\|_{L^\infty}^2,$$

where C is a constant that depends on the systems initial conditions and the Reynolds number.

Existence of an Optimal ϵ

To ensure smooth solutions, we need to find a lower bound on ϵ such that the regularization remains effective. We establish that for any initial condition, there exists an optimal ϵ such that:

$$\epsilon_{\text{opt}} \leq \epsilon < \infty,$$

where ϵ_{opt} is a function of the initial energy $E(0)$ and the initial velocity gradients. The optimal value of ϵ ensures that dissipation dominates energy concentration throughout the evolution of the system.

Asymptotic Behavior as $\epsilon \rightarrow 0$

As ϵ approaches zero, the system approaches the classical Navier-Stokes equations. However, for any physically relevant timescale, we can choose ϵ such that the solution remains smooth. Specifically, as $\epsilon \rightarrow 0$, the regularization term diminishes:

$$\lim_{\epsilon \rightarrow 0} \epsilon \mathbf{R}(\mathbf{u}) = 0,$$

but the velocity gradients remain bounded due to the residual effect of viscosity:

$$\frac{dE(t)}{dt} = -\nu \|\nabla \mathbf{u}\|_{L^2}^2.$$

This ensures that the system remains well-behaved for small values of ϵ , with the viscosity providing enough dissipation to prevent singularities for physically relevant timescales.

Global Bound for $\epsilon > 0$

For small but nonzero ϵ , we can establish a global bound on the velocity field. From the energy inequality:

$$E(t) + \nu \int_0^T \|\nabla \mathbf{u}\|_{L^2}^2 dt + \epsilon \int_0^T \|\nabla \mathbf{u}\|_{L^2}^2 dt \leq E(0),$$

we conclude that the solution remains globally bounded for all time, even as $\epsilon \rightarrow 0$.

Thus, we have shown that the morphic-regularized Navier-Stokes equations remain smooth for small ϵ , with the solution approaching the classical system as $\epsilon \rightarrow 0$, while still ensuring smoothness for physically relevant timescales.

2 Conclusion of Step 4

We have demonstrated that the regularization parameter ϵ can be chosen to balance smoothness and accuracy. Even as $\epsilon \rightarrow 0$, the system remains smooth for physically relevant timescales. This establishes that the morphic-regularized Navier-Stokes equations can handle the small ϵ limit while preventing singularities.

Next, we will address ****Step 5: Uniqueness of Solutions****.

Introduction

In this final step, we address the uniqueness of solutions to the morphic-regularized Navier-Stokes equations. While the introduction of the regularization term $\epsilon \mathbf{R}(\mathbf{u})$ provides smooth solutions, we now focus on proving that these solutions are unique, ensuring that the system yields a single, well-defined solution for any given set of initial conditions.

Step 5: Uniqueness of Solutions

The uniqueness of the Navier-Stokes solutions is a central challenge. To prove uniqueness for the morphic-regularized system, we must show that if two solutions exist, their difference diminishes over time, ultimately converging to zero. This step involves comparing two potential solutions $\mathbf{u}_1$ and $\mathbf{u}_2$, and demonstrating that the difference between them vanishes.

Consider Two Solutions

Let $\mathbf{u}_1(x, t)$ and $\mathbf{u}_2(x, t)$ be two smooth solutions to the morphic-regularized Navier-Stokes equations. We define their difference as:

$$\mathbf{w}(x, t) = \mathbf{u}_1(x, t) - \mathbf{u}_2(x, t).$$

Taking the difference between the two versions of the Navier-Stokes equations gives:

$$\frac{\partial \mathbf{w}}{\partial t} + (\mathbf{u}_1 \cdot \nabla) \mathbf{w} - (\mathbf{u}_2 \cdot \nabla) \mathbf{w} = \nu \nabla^2 \mathbf{w} + \epsilon \mathbf{R}(\mathbf{w}).$$

Since $\mathbf{u}_1$ and $\mathbf{u}_2$ are both incompressible, we also have:

$$\nabla \cdot \mathbf{w} = 0.$$

Energy Estimate for $\mathbf{w}(x, t)$

We now derive an energy estimate for $\mathbf{w}(x, t)$. Taking the L^2 -norm of the difference $\mathbf{w}$ yields the energy:

$$E_w(t) = \frac{1}{2} \int_{\mathbb{R}^n} |\mathbf{w}(x, t)|^2 dx.$$

Taking the time derivative of $E_w(t)$ gives:

$$\frac{d}{dt} E_w(t) = \int_{\mathbb{R}^n} \mathbf{w}(x, t) \cdot \frac{\partial \mathbf{w}}{\partial t} dx.$$

Substituting the equation for $\frac{\partial \mathbf{w}}{\partial t}$, we get:

$$\frac{d}{dt} E_w(t) = \int_{\mathbb{R}^n} \mathbf{w} \cdot (-(\mathbf{u}_1 \cdot \nabla) \mathbf{w} + (\mathbf{u}_2 \cdot \nabla) \mathbf{w} + \nu \nabla^2 \mathbf{w} + \epsilon \mathbf{R}(\mathbf{w})) dx.$$

The convective terms cancel out due to incompressibility, and the remaining terms yield:

$$\frac{d}{dt} E_w(t) = -\nu \|\nabla \mathbf{w}\|_{L^2}^2 + \epsilon \int_{\mathbb{R}^n} \mathbf{w} \cdot \mathbf{R}(\mathbf{w}) dx.$$

Dissipation of Energy Difference

The viscous term $-\nu \|\nabla \mathbf{w}\|_{L^2}^2$ ensures dissipation of the energy difference over time. The regularization term $\epsilon \mathbf{R}(\mathbf{w})$ provides additional smoothing, dissipating any remaining energy in $\mathbf{w}$. Since the regularization term is dissipative, we have:

$$\int_{\mathbb{R}^n} \mathbf{w} \cdot \mathbf{R}(\mathbf{w}) dx \leq 0,$$

which guarantees further dissipation.

Thus, the total dissipation is given by:

$$\frac{d}{dt} E_w(t) \leq -\nu \|\nabla \mathbf{w}\|_{L^2}^2.$$

This implies that the energy difference $E_w(t)$ decreases over time, driving $\mathbf{w}(x, t)$ toward zero.

Uniqueness of Solutions

Since $E_w(t) \geq 0$ and $\frac{d}{dt}E_w(t) \leq 0$, it follows that $E_w(t) \rightarrow 0$ as $t \rightarrow \infty$. Therefore, the difference between the two solutions $\mathbf{u}_1$ and $\mathbf{u}_2$ vanishes over time, meaning that:

$$\mathbf{u}_1(x, t) = \mathbf{u}_2(x, t) \quad \text{for all } t.$$

This establishes that the morphic-regularized Navier-Stokes equations have a unique solution for all initial conditions.

3 Conclusion of Step 5

We have demonstrated that the morphic-regularized Navier-Stokes equations admit a unique smooth solution for all time. The dissipation introduced by the regularization term ensures that any two solutions will converge to the same result, proving uniqueness.